

Basketball Dice v0.4

Possession Procedure Guide

This replaces the old possession procedure section. It is written as a tabletop checklist: do the steps in order, stop when the possession ends, and only loop when the offense gets an offensive rebound.

Important: a possession is counted once at the start. A non-shooting foul does not end the possession. An offensive rebound does not create a new possession.

You need	Why
Two d10s / d100	Every roll is 01-100. Treat 00 as 100.
Team card	Team modifiers for defense, shooting quality, fouls, threes, rebounds, assists.
Player card	The selected player's usage, shooting, FT, turnover, foul-drawn, and weights.
Use and event tables	Use table picks action player. Event tables assign AST, REB, STL, BLK, PF.
Scoresheet	Track player stats and team totals.

The possession loop

Follow this sequence. When a branch says "possession ends," stop and give the ball to the other team. When a branch says "loop," keep the same possession and return to the loose foul check.

Step	Name	What to do
1	Mark possession	Add one possession for the offense. Do this only once.
2	Loose foul check	Roll d100. 01-07 = defensive non-shooting foul. Assign PF, then continue. 08-100 = no foul.
3	Pick action player	Roll d100 on the offense Use table. This player will turn it over, draw a foul, or shoot.
4	Action roll	Calculate TOV and FD. Roll d100. Low roll = turnover, middle = shooting foul, high = shot attempt.
5A	Turnover branch	Record TOV. Roll steal check. Maybe assign STL. Possession ends.
5B	Shooting foul branch	Assign defender PF. Shoot 2 free throws. Possession ends.
5C	Shot branch	Roll shot type, then make/miss. Made shot can get an assist and ends. Miss goes to block/rebound.
6	Rebound branch	If offense rebounds, assign OREB and loop to Step 2 without a new possession. Otherwise assign DREB and end.

End conditions: turnover, shooting foul free throws, made field goal, defensive rebound.

Continue conditions: non-shooting foul, offensive rebound.

Step-by-step details

Step 1 - Mark possession

Add 1 team possession for the offense. For Thunder at Knicks, use 98 possessions per team: 25 in Q1, 25 in Q2, 24 in Q3, and 24 in Q4. Alternate away and home possessions until both teams have used the quarter total.

Step 2 - Loose / non-shooting foul check

Roll d100. On 01-07, the defense commits a non-shooting foul. Assign the PF using the defense PF table. Record that player PF +1 and team PF +1. Then continue. No free throws are awarded for these fouls in v0.4.

Step 3 - Pick the action player

Roll d100 on the offense Use table. The selected player is the action player. Use him for the turnover, foul-drawn, shot-type, shooting, and FT numbers if needed.

Step 4 - Action roll

Number	Formula
Effective TOV	Player TOV + defense TO Press - offense TO Protect
Effective FD	Player FD + offense Foul Draw - defense Foul Disc + 1

After you calculate those two numbers, roll d100:

Roll range	Result
01 through Effective TOV	Turnover
Next Effective FD numbers	Shooting foul
Anything higher	Shot attempt

Branch 5A - Turnover

If the action roll is in the turnover range:

Order	Do this
1	Record TOV +1 for the action player and team TOV +1.
2	Roll d100 for the steal check.
3	01-60 = defense gets a steal. 61-100 = no steal.
4	If there is a steal, roll on the defense STL table and record STL +1.
5	Possession ends.

Branch 5B - Shooting foul

Order	Do this
1	Assign the defensive foul using the defense PF table. Record PF +1 for that player and team PF +1.
2	The action player shoots 2 free throws.
3	Roll d100 for each free throw. A roll <= player FT is made.
4	Record FTA, FTM, points, and team points. Do not record an FGA.
5	Possession ends.

Branch 5C - Shot attempt

First determine whether the shot is a three.

Question	Formula
3PT chance	Player 3F + offense 3PT Tend

Roll d100. If the roll is $\leq$ the 3PT chance, record it as a 3PA. Otherwise it is a 2PA.

Now determine whether the shot is made.

Shot	Make formula
2PA	Player 2P + offense ShotQ - floor(defense DEF / 2) - 1
3PA	Player 3P + offense ShotQ - floor(defense DEF / 2) - 1

Roll d100. If the roll is $\leq$ the make number, the shot is made. Otherwise it is missed.

On every shot attempt: record FGA. If it was a three, also record 3PA. If made, record FGM, points, and 3PM if it was a three.

If the shot is made: assist check

Made shot	Assist chance
Made 2	Team AST
Made 3	Team AST + 8

If the assist roll succeeds, roll on the offense AST table. If the table selects the shooter, reroll until a teammate is selected. Record AST +1 for that teammate. The possession ends.

If the shot is missed: block and rebound

Only missed 2PA can be blocked in this version.

Check	Formula / result
Block chance	7 + defense DEF
If block succeeds	Assign BLK using the defense BLK table.
If block fails	Continue to rebound.

Then resolve the rebound:

Check	Formula / result
Offensive rebound chance	27 + offense ORB - defense DRB
If offense rebounds	Assign OREB using offense OREB table. Do not add a possession. Return to Step 2.
If defense rebounds	Assign DREB using defense DREB table. Possession ends.

A single possession can be extended by offensive rebounds at most 2 times. After 2 offensive rebound extensions, the next miss goes to the defense automatically.

Worked example 1 - made assisted three

#	Play-by-play
1	Thunder offense. Mark OKC POSS +1.
2	Loose foul roll = 46. No foul.
3	Use roll = 17. On the Thunder Use table, 01-27 = Shai Gilgeous-Alexander.
4	Effective TOV = Shai 8 + Knicks TO Press 1 - Thunder TO Protect 1 = 8.
5	Effective FD = Shai 15 + Thunder Foul Draw 1 - Knicks Foul Disc 0 + 1 = 17.
6	Action roll = 42. Turnover is 01-08. Shooting foul is 09-25. So 42 = shot attempt.
7	3PT chance = Shai 3F 26 + Thunder 3PT Tend 1 = 27. Shot type roll = 19, so it is a 3PA.
8	3P make = Shai 37 + Thunder ShotQ 1 - floor(Knicks DEF 1 / 2) 0 - 1 = 37. Make roll = 34, so it is good.
9	Assist chance = Thunder AST 57 + 8 because it was a made three = 65. Assist roll = 51, so assisted. Roll on Thunder AST table, rerolling if Shai is selected.
10	Record Shai FGA +1, 3PA +1, FGM +1, 3PM +1, PTS +3. Record passer AST +1. Thunder score +3. Possession ends.

Worked example 2 - miss, offensive rebound, same possession continues

#	Play-by-play
1	Knicks offense. Mark NYK POSS +1.
2	Loose foul roll = 33. No foul.
3	Use roll = 45. On the Knicks Use table, 28-49 = Karl-Anthony Towns.
4	Effective TOV = Towns 11 + Thunder TO Press 2 - Knicks TO Protect 1 = 12.
5	Effective FD = Towns 10 + Knicks Foul Draw 1 - Thunder Foul Disc 1 + 1 = 11.
6	Action roll = 58. Turnover is 01-12. Shooting foul is 13-23. So 58 = shot attempt.
7	3PT chance = Towns 38 + Knicks 3PT Tend 1 = 39. Shot type roll = 92, so it is a 2PA.
8	2P make = Towns 58 + Knicks ShotQ 1 - floor(Thunder DEF 5 / 2) 2 - 1 = 56. Make roll = 67, so it misses. Record Towns FGA +1.
9	Block chance = 7 + Thunder DEF 5 = 12. Block roll = 40, so no block.
10	Offensive rebound chance = 27 + Knicks ORB 2 - Thunder DRB 2 = 27. Rebound roll = 14, so Knicks get an OREB.
11	Assign OREB using Knicks OREB table. Do not add a new possession. Return to Step 2 and keep playing this Knicks possession.

Worked example 3 - turnover with steal

#	Play-by-play
1	Thunder offense. Mark OKC POSS +1.
2	Loose foul roll = 81. No foul.
3	Use roll = 39. On the Thunder Use table, 28-45 = Jalen Williams.
4	Effective TOV = Jalen 10 + Knicks TO Press 1 - Thunder TO Protect 1 = 10.
5	Effective FD = Jalen 7 + Thunder Foul Draw 1 - Knicks Foul Disc 0 + 1 = 9.
6	Action roll = 07. That is inside 01-10, so it is a turnover.
7	Record Jalen Williams TOV +1 and Thunder team TOV +1.
8	Steal check roll = 44. Since $44 \leq 60$, the Knicks get a steal. Assign STL using Knicks STL table. Possession ends.

Event assignment tables

When the rules say "assign" a stat, use these d100 tables. This avoids recalculating player weights during a game.

Event	Use this team	Rule
Use	Offense	Picks the action player.
AST	Offense	If the shooter is selected, reroll until a teammate is selected.
OREB	Offense	Used only after an offensive rebound result.
DREB	Defense	Used only after the offense fails the offensive rebound roll.
STL	Defense	Used only when turnover steal check succeeds.
BLK	Defense	Used only when missed-2 block check succeeds.
PF	Defense	Used for loose fouls and shooting fouls.

Thunder d100 assignment tables

Pace	ShotQ	DEF	Press	Protect	Draw	Disc	3PT	ORB	DRB	AST
100.0	1	5	2	1	1	1	1	0	2	57

Use	Player
01-27	Shai Gilgeous-Alexander
28-45	Jalen Williams
46-57	Chet Holmgren
58-65	Luguentz Dort
66-72	Isaiah Hartenstein
73-79	Cason Wallace
80-84	Alex Caruso
85-91	Isaiah Joe
92-100	Aaron Wiggins

OREB	Player
01-04	Shai Gilgeous-Alexander
05-12	Jalen Williams
13-35	Chet Holmgren
36-45	Luguentz Dort
46-76	Isaiah Hartenstein
77-82	Cason Wallace
83-88	Alex Caruso
89-91	Isaiah Joe
92-100	Aaron Wiggins

STL	Player
01-18	Shai Gilgeous-Alexander
19-31	Jalen Williams
32-39	Chet Holmgren
40-52	Luguentz Dort
53-57	Isaiah Hartenstein
58-70	Cason Wallace
71-85	Alex Caruso
86-92	Isaiah Joe
93-100	Aaron Wiggins

PF	Player
01-11	Shai Gilgeous-Alexander
12-22	Jalen Williams
23-35	Chet Holmgren
36-49	Luguentz Dort
50-66	Isaiah Hartenstein
67-75	Cason Wallace
76-85	Alex Caruso
86-91	Isaiah Joe
92-100	Aaron Wiggins

AST	Player
01-29	Shai Gilgeous-Alexander
30-48	Jalen Williams
49-56	Chet Holmgren
57-62	Luguentz Dort
63-73	Isaiah Hartenstein
74-80	Cason Wallace
81-89	Alex Caruso
90-93	Isaiah Joe
94-100	Aaron Wiggins

DREB	Player
01-10	Shai Gilgeous-Alexander
11-22	Jalen Williams
23-42	Chet Holmgren
43-52	Luguentz Dort
53-74	Isaiah Hartenstein
75-80	Cason Wallace
81-87	Alex Caruso
88-91	Isaiah Joe
92-100	Aaron Wiggins

BLK	Player
01-11	Shai Gilgeous-Alexander
12-20	Jalen Williams
21-62	Chet Holmgren
63-68	Luguentz Dort
69-83	Isaiah Hartenstein
84-87	Cason Wallace
88-91	Alex Caruso
92-94	Isaiah Joe
95-100	Aaron Wiggins

Knicks d100 assignment tables

Pace	ShotQ	DEF	Press	Protect	Draw	Disc	3PT	ORB	DRB	AST
96.7	1	1	1	1	1	0	1	2	0	56

Use	Player
01-27	Jalen Brunson
28-49	Karl-Anthony Towns
50-65	Mikal Bridges
66-80	OG Anunoby
81-89	Josh Hart
90-94	Miles McBride
95-98	Mitchell Robinson
99-100	Precious Achiuwa

OREB	Player
01-02	Jalen Brunson
03-24	Karl-Anthony Towns
25-29	Mikal Bridges
30-37	OG Anunoby
38-52	Josh Hart
53-56	Miles McBride
57-84	Mitchell Robinson
85-100	Precious Achiuwa

STL	Player
01-13	Jalen Brunson
14-22	Karl-Anthony Towns
23-41	Mikal Bridges
42-61	OG Anunoby
62-75	Josh Hart
76-85	Miles McBride
86-93	Mitchell Robinson
94-100	Precious Achiuwa

PF	Player
01-12	Jalen Brunson
13-29	Karl-Anthony Towns
30-41	Mikal Bridges
42-56	OG Anunoby
57-69	Josh Hart
70-75	Miles McBride
76-90	Mitchell Robinson
91-100	Precious Achiuwa

AST	Player
01-34	Jalen Brunson
35-49	Karl-Anthony Towns
50-62	Mikal Bridges
63-71	OG Anunoby
72-86	Josh Hart
87-94	Miles McBride
95-97	Mitchell Robinson
98-100	Precious Achiuwa

DREB	Player
01-06	Jalen Brunson
07-30	Karl-Anthony Towns
31-39	Mikal Bridges
40-49	OG Anunoby
50-66	Josh Hart
67-71	Miles McBride
72-89	Mitchell Robinson
90-100	Precious Achiuwa

BLK	Player
01-03	Jalen Brunson
04-25	Karl-Anthony Towns
26-32	Mikal Bridges
33-41	OG Anunoby
42-46	Josh Hart
47-49	Miles McBride
50-84	Mitchell Robinson
85-100	Precious Achiuwa

Celtics d100 assignment tables

Pace	ShotQ	DEF	Press	Protect	Draw	Disc	3PT	ORB	DRB	AST
95.7	2	3	1	1	-1	1	4	-1	1	62

Use	Player
01-25	Jayson Tatum
26-47	Jaylen Brown
48-62	Derrick White
63-76	Kristaps Porzingis
77-85	Jrue Holiday
86-92	Payton Pritchard
93-96	Al Horford
97-99	Sam Hauser
100-100	Luke Kornet

OREB	Player
01-14	Jayson Tatum
15-25	Jaylen Brown
26-32	Derrick White
33-50	Kristaps Porzingis
51-59	Jrue Holiday
60-66	Payton Pritchard
67-77	Al Horford
78-82	Sam Hauser
83-100	Luke Kornet

STL	Player
01-14	Jayson Tatum
15-29	Jaylen Brown
30-44	Derrick White
45-50	Kristaps Porzingis
51-71	Jrue Holiday
72-82	Payton Pritchard
83-90	Al Horford
91-96	Sam Hauser
97-100	Luke Kornet

PF	Player
01-11	Jayson Tatum
12-23	Jaylen Brown
24-34	Derrick White
35-48	Kristaps Porzingis
49-59	Jrue Holiday
60-67	Payton Pritchard
68-79	Al Horford
80-86	Sam Hauser
87-100	Luke Kornet

AST	Player
01-20	Jayson Tatum
21-33	Jaylen Brown
34-51	Derrick White
52-58	Kristaps Porzingis
59-74	Jrue Holiday
75-85	Payton Pritchard
86-93	Al Horford
94-97	Sam Hauser
98-100	Luke Kornet

DREB	Player
01-18	Jayson Tatum
19-31	Jaylen Brown
32-39	Derrick White
40-56	Kristaps Porzingis
57-66	Jrue Holiday
67-72	Payton Pritchard
73-85	Al Horford
86-91	Sam Hauser
92-100	Luke Kornet

BLK	Player
01-09	Jayson Tatum
10-16	Jaylen Brown
17-31	Derrick White
32-60	Kristaps Porzingis
61-66	Jrue Holiday
67-69	Payton Pritchard
70-82	Al Horford
83-84	Sam Hauser
85-100	Luke Kornet

Nuggets d100 assignment tables

Pace	ShotQ	DEF	Press	Protect	Draw	Disc	3PT	ORB	DRB	AST
99.8	2	-1	0	2	1	-1	-1	0	0	66

Use	Player
01-29	Nikola Jokic
30-53	Jamal Murray
54-69	Michael Porter Jr.
70-81	Aaron Gordon
82-89	Christian Braun
90-95	Russell Westbrook
96-98	Peyton Watson
99-100	Julian Strawther

OREB	Player
01-29	Nikola Jokic
30-34	Jamal Murray
35-46	Michael Porter Jr.
47-66	Aaron Gordon
67-76	Christian Braun
77-86	Russell Westbrook
87-95	Peyton Watson
96-100	Julian Strawther

STL	Player
01-16	Nikola Jokic
17-30	Jamal Murray
31-40	Michael Porter Jr.
41-52	Aaron Gordon
53-67	Christian Braun
68-84	Russell Westbrook
85-94	Peyton Watson
95-100	Julian Strawther

PF	Player
01-14	Nikola Jokic
15-24	Jamal Murray
25-36	Michael Porter Jr.
37-52	Aaron Gordon
53-66	Christian Braun
67-80	Russell Westbrook
81-94	Peyton Watson
95-100	Julian Strawther

AST	Player
01-36	Nikola Jokic
37-56	Jamal Murray
57-60	Michael Porter Jr.
61-70	Aaron Gordon
71-77	Christian Braun
78-93	Russell Westbrook
94-97	Peyton Watson
98-100	Julian Strawther

DREB	Player
01-31	Nikola Jokic
32-40	Jamal Murray
41-55	Michael Porter Jr.
56-68	Aaron Gordon
69-77	Christian Braun
78-89	Russell Westbrook
90-96	Peyton Watson
97-100	Julian Strawther

BLK	Player
01-22	Nikola Jokic
23-28	Jamal Murray
29-40	Michael Porter Jr.
41-54	Aaron Gordon
55-63	Christian Braun
64-69	Russell Westbrook
70-96	Peyton Watson
97-100	Julian Strawther